

Efficient Trip Planning

A group of friends want to plan a trip in a foreign country and want to make sure to visit all the interesting place in the most efficient manner. To do this, they start listing the road they will take to reach each places. The important part here is that they don't wanna go to the place that they have visited and take the same road (a cycle) for time and fuel efficiency.

Input Format

- The first line contains integers P and R , where P is the number of place and R is the total number of roads connecting the places. The roads are bidirectional.
- The next R lines contains a and b (the place ids), indicating the place has a road connecting them.

Output Format

Whether the planning have a cycle or not (SAFE/CYCLE).

Constraints

- $1 \leq a, b \leq C$
- $2 \leq C \leq 100$
- $1 \leq R \leq (C * (C - 1))/2$
- Implement your algorithm in C using Disjoint Set concept.

Sample Input 1 (standard input)

```
6 5
1 2
2 3
3 6
6 5
5 4
```

Sample Output 1 (standard output)

```
SAFE
```

Explanation 1

The plan is perfect already, they don't take any road that they have taken and don't visit any visited places.

Sample Input 2 (standard input)

```
4 3
2 3
3 1
1 2
```

Sample Output 2 (standard output)

```
CYCLE
```

Explanation 2

Road 1 2 makes a cycle in the whole plan because they have visited 2 before.